

Presentation to begin

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Tyler Capital



Machine Learning & Trading



Tyler Capital is a 15-year-old London based trading firm, focussed on the systematic application of machine learning to the global financial markets. Tyler Capital sits at the apex of high frequency trading and artificial intelligence. The firm builds sophisticated machine learning trading systems using some of the most advanced AI techniques in the world. As well, Tyler has developed a unique manufacturing process for the development of AI-based trading systems that ensures they remain, competitive, compliant and above all ethical.



www.tylercapital.co.uk



[linkedin.com/company/tyler-capital](https://www.linkedin.com/company/tyler-capital)



15 St Botolph St, London, EC3A 7DT



020.7663.3700



About me

I'm smart enough to know that I'm dumb. - *Richard Feynman*

The way you become an expert in a field is to make every mistake possible in that field. - *Niels Bohr*

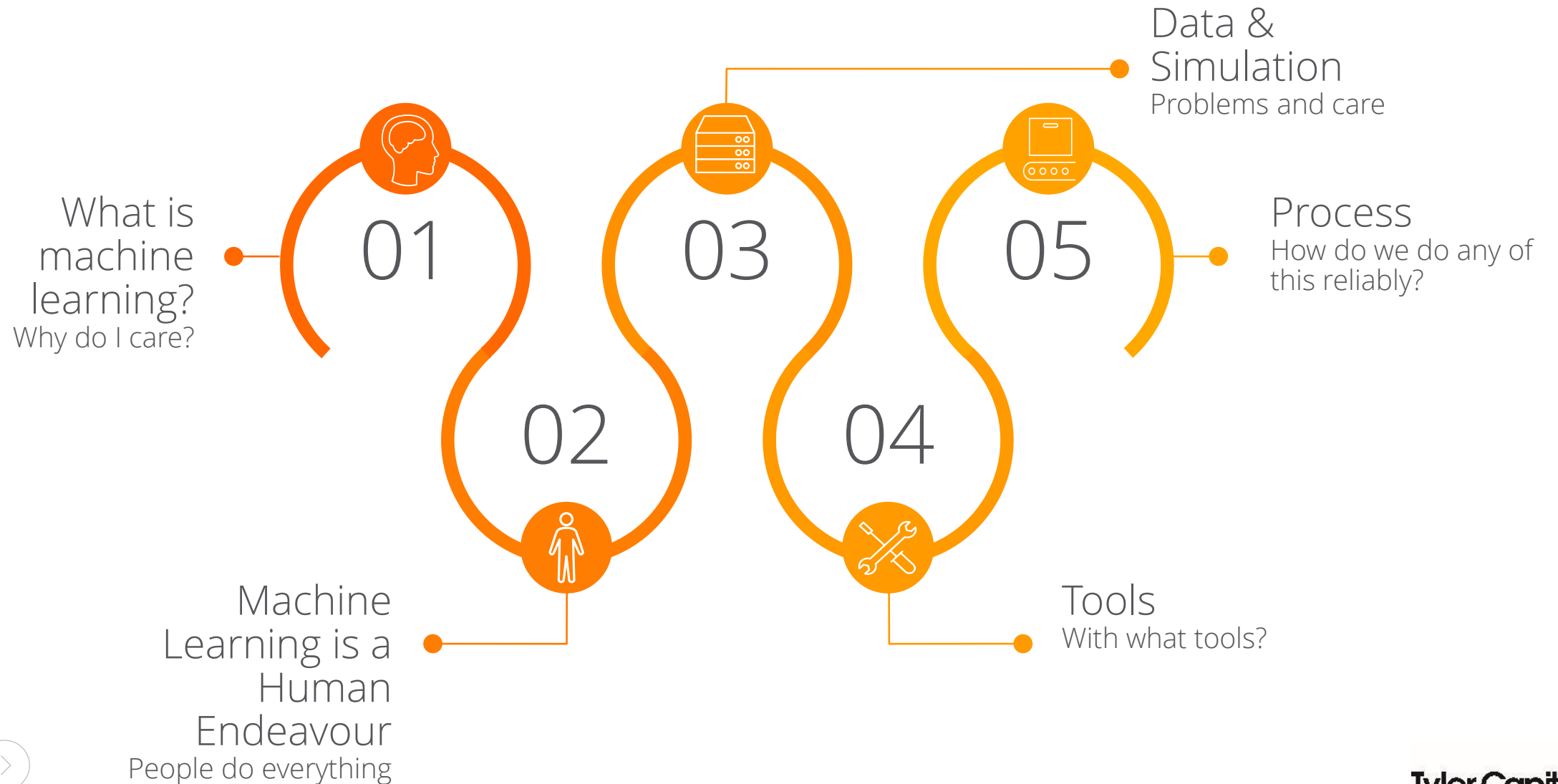
Disclaimers

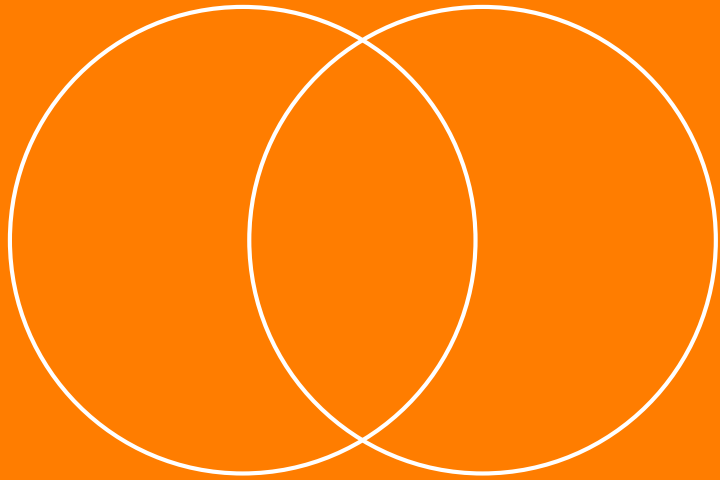


At Tyler Capital, we do this for a living. There will be no “secrets” today. Everything we will discuss is available in the public domain. It will still be useful.

This content an opinion and is for information purposes only. It is not intended to be investment advice. Carry out your own independent research. Carry out your own independent verification of facts and data. Seek professional advice if required.

Outline





What is machine learning?

Machine Learning is a Human Endeavour

- AI is a human endeavour - it is a different kind of technology, where you must serve it collaboratively. It harnesses your intelligences. It learns on its own and will move on beyond you.
- It can't serve you, you must serve it. It is not 'your quant' - it is something that has its own goals, biases, etc. You have to INFLUENCE your AI.

Lead	Empathize	Create	Judge	Train	Explain	Sustain	Amplify	Interact	Embody	Transact	Iterate	Predict	Adapt
H Human-only activity				Humans complement machines		AI gives humans superpowers			M Machine-only activity				
				Human and machine hybrid activities									

THIS IS YOUR MACHINE LEARNING SYSTEM?

YUP! YOU POUR THE DATA INTO THIS BIG
PILE OF LINEAR ALGEBRA, THEN COLLECT
THE ANSWERS ON THE OTHER SIDE.

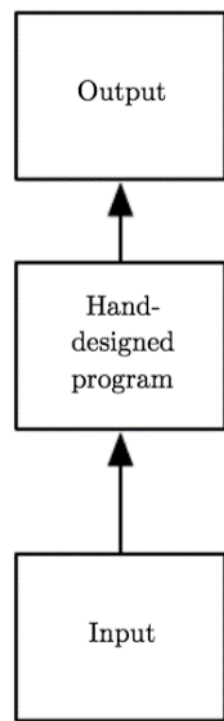
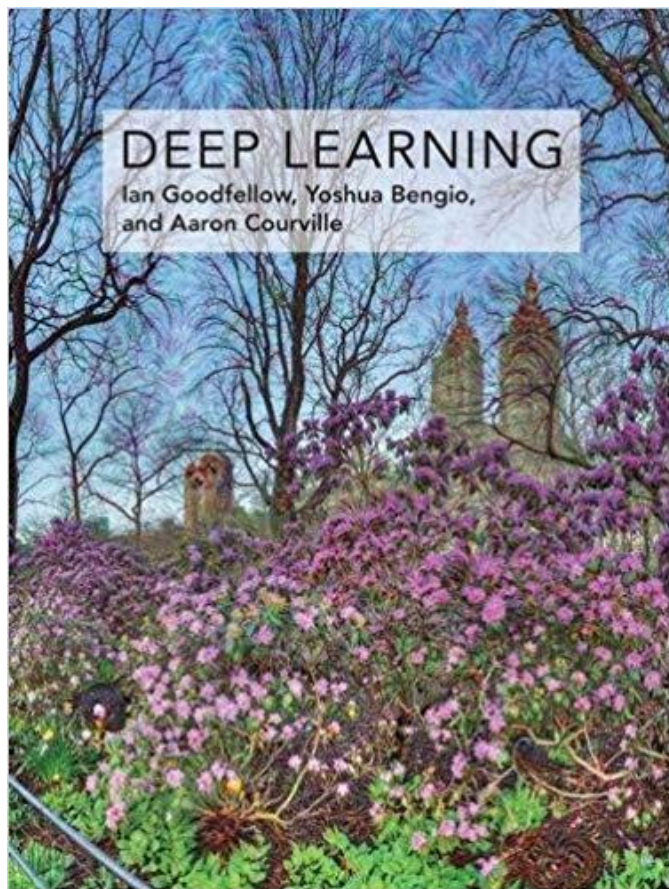
WHAT IF THE ANSWERS ARE WRONG?

JUST STIR THE PILE UNTIL
THEY START LOOKING RIGHT.

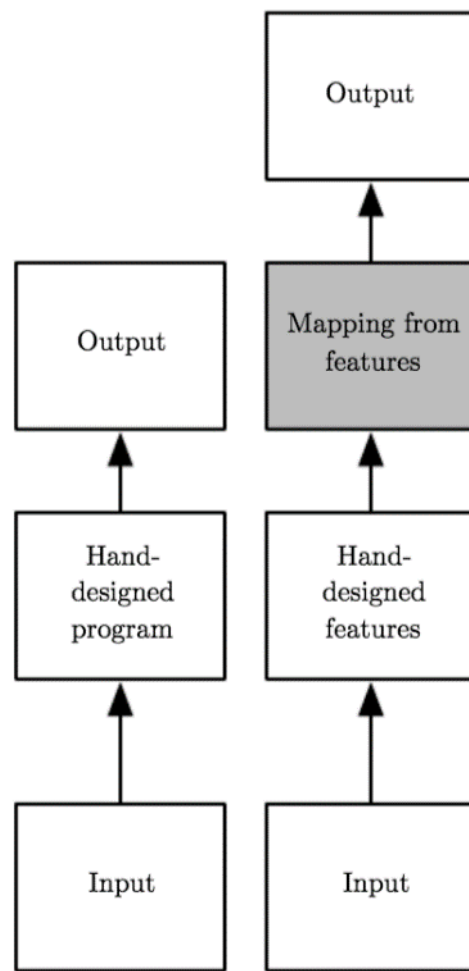
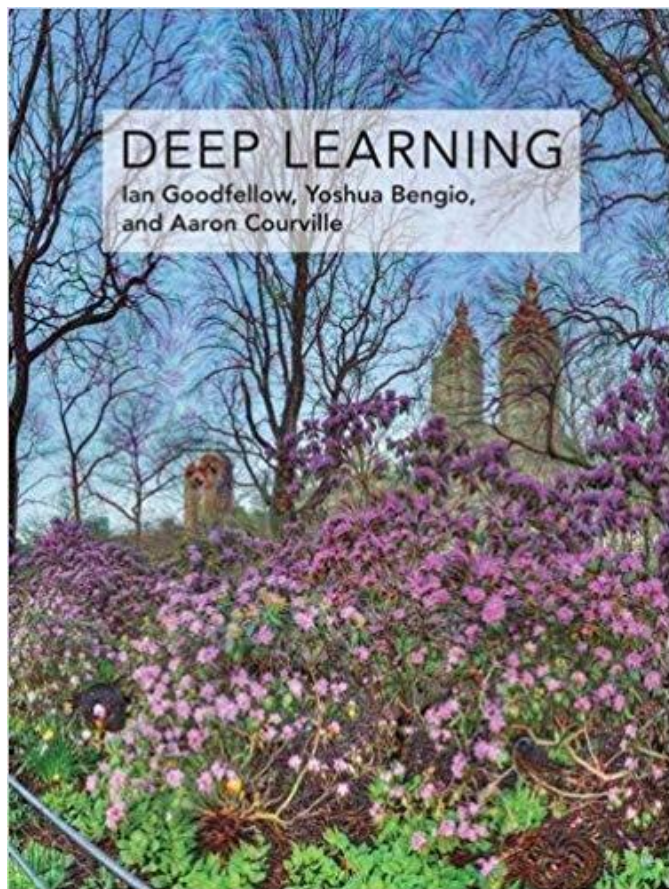


<https://xkcd.com/1838/>

Tyler Capital 

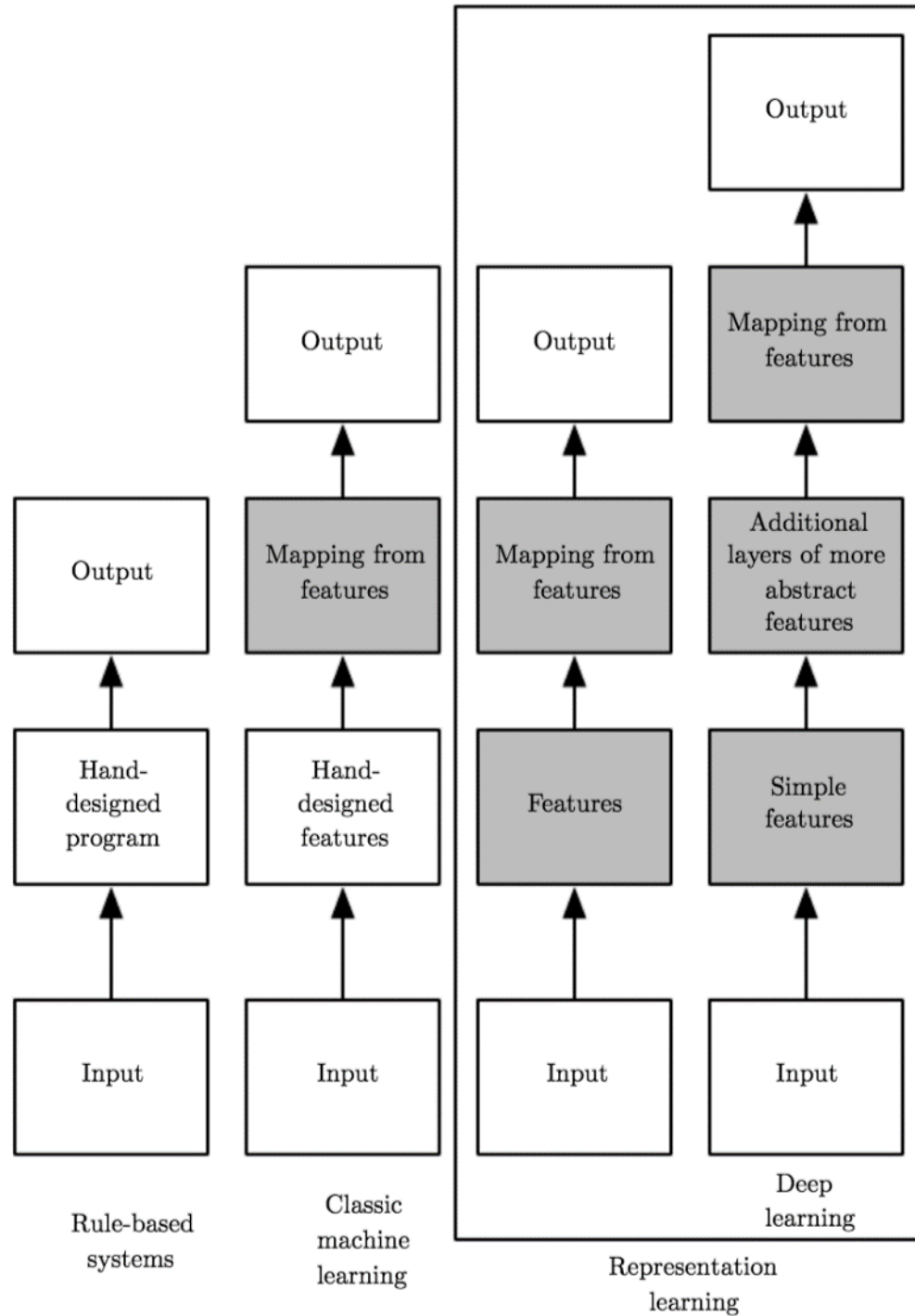
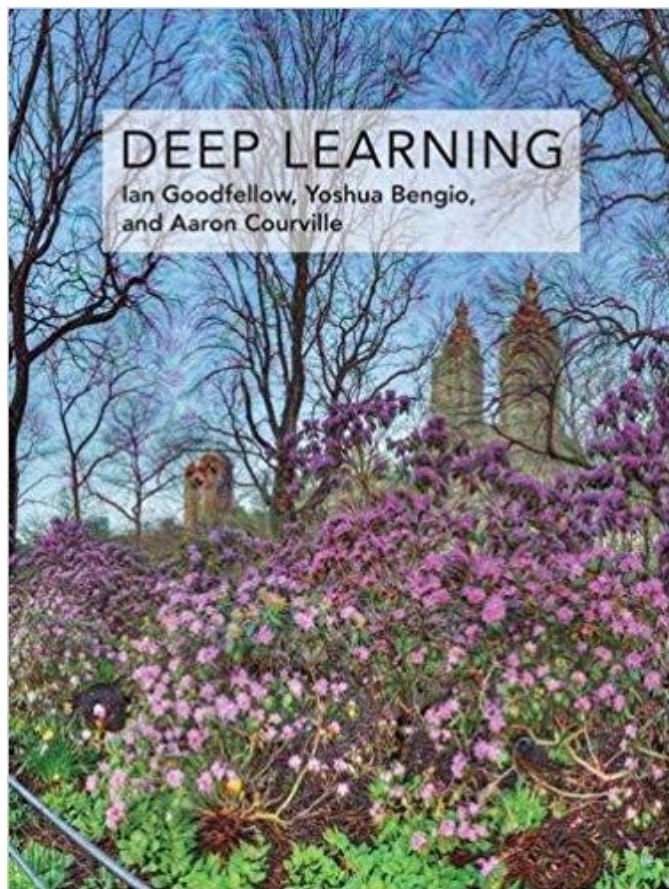


Rule-based
systems



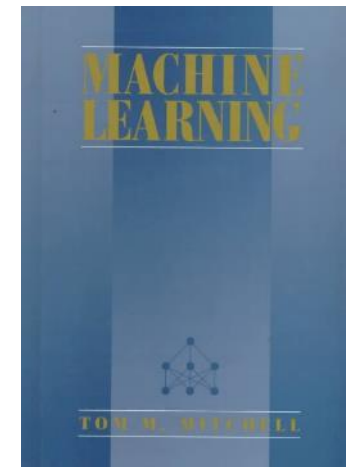
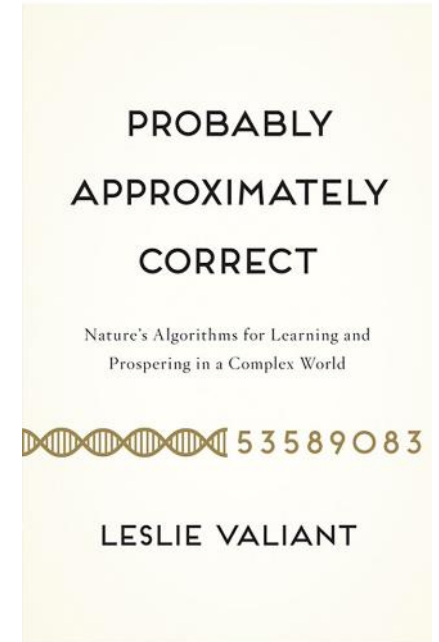
Rule-based
systems

Classic
machine
learning



Other Useful Definitions

- Machine learning, a method for finding patterns in data that are usefully predictive of future events but which do not necessarily provide an explanatory theory. [Valiant]
- Machine learning is the general field that studies how complex mechanisms can be created without a designer. [Valiant]
- Machine learning: How to construct programs that automatically learn from experience [Mitchell]
- The goal of machine learning is to work on unseen data, not seen data.

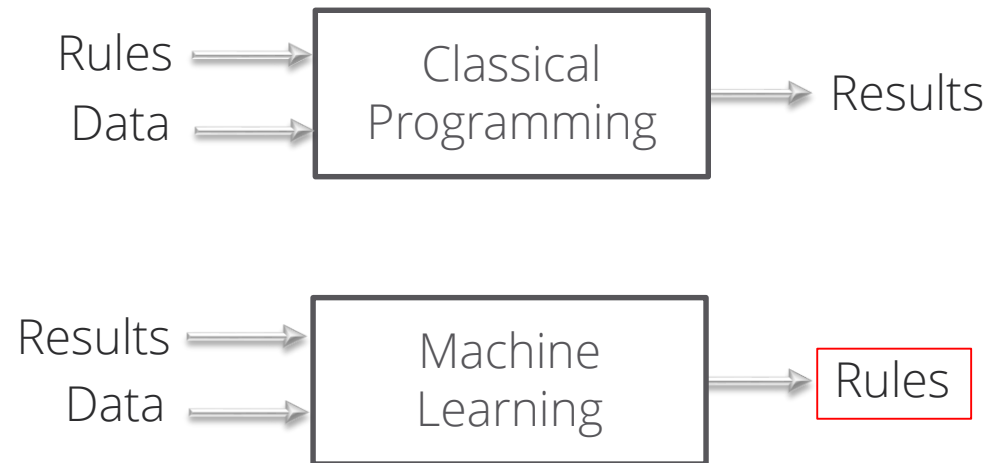


The Traditional Way to Compete



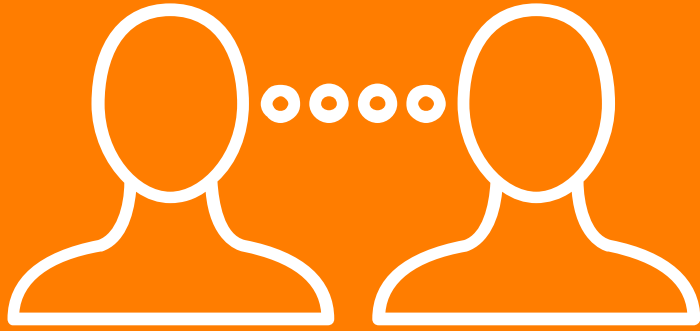
Decay
Keep making rules or die.

Adaptation



Adaptive

An algorithm that changes its behavior, based on information available and an a priori defined reward mechanism.



Machine Learning is a
Human Endeavor

Marcos Lopez De Prado Wisdom



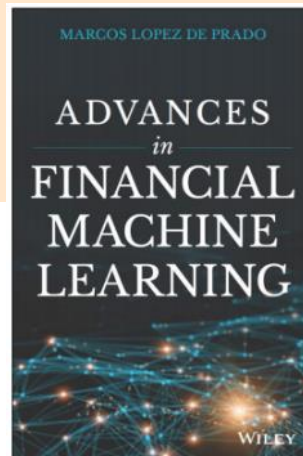
Key Challenge

- ML experts have little experience with trading
- Trading experts have little experience with machine learning



Important Myth

- ML is a black box.
- You can explain how ML systems work



Meta-Strategy Paradigm

- Silos and isolation will not work
- Collaboration needed

The Meta-Strategy Paradigm

- The complexities involved in developing a true investment strategy are overwhelming:
 - Data collection, curation, processing, structuring,
 - HPC infrastructure,
 - software development,
 - feature analysis,
 - execution simulators,
 - backtesting, etc.
- Even if the firm provides you with shared services in those areas, you are like a worker at a BMW factory who has been asked to build the entire car alone, by using all the workshops around you.
 - One week you need to be a master welder, another week an electrician, another week a mechanical engineer, another week a painter, ... try, fail and circle back to welding. It is a futile endeavor.

Marcos López de Prado

The Meta-Strategy Paradigm

- It takes almost as much effort to produce one true investment strategy as to produce a hundred.
- Every successful quantitative firm I am aware of applies the [meta-strategy paradigm](#).
- Your firm must set up a research factory
 - where tasks of the assembly line are clearly divided into subtasks.
 - where quality is independently measured and monitored for each subtask.
 - where the role of each quant is to specialize in a particular subtask, to become the best there is at it, while having a holistic view of the entire process.
- This is how Berkeley Lab and other U.S. National laboratories routinely make scientific discoveries, such as adding 16 elements to the periodic table, or laying out the groundwork for MRIs and PET scans: <https://youtu.be/G5nK3B5uuY8>

Marcos López de Prado

Jobs for trainers

RELATIONSHIP ARCHITECTURE



Human *actively* trains agent for task performance and/or humanness.



Crowd of humans *actively* trains agent to improve task performance and/or humanness.



Crowd of humans *passively* trains agent to boost task performance and/or humanness, while the robot progressively offloads elements of the task.

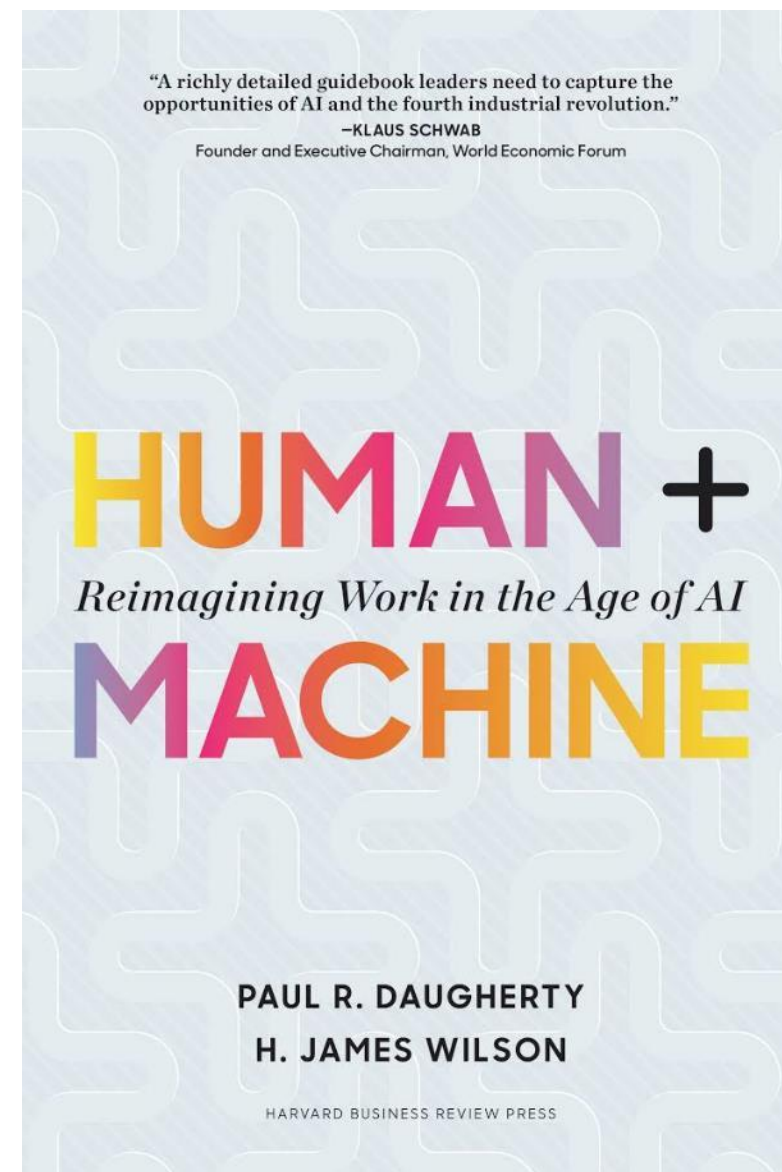
EXAMPLE ACTIVITIES

Task performance training:

- “Clean” data for upload
- Discover relevant data and data streams
- Have machine observe decision making
- Tag data for better use
- Work with HR to inform the design of workplace retraining initiatives

Humanness attribute training:

- Train language, gesture, empathy
- Have machine observe interactions
- Correct errors, reinforce successes
- Define and develop personality

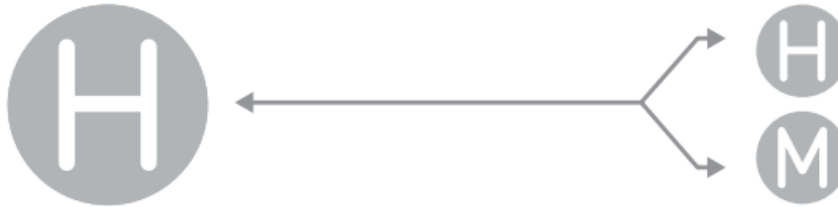


Jobs for explainers

RELATIONSHIP ARCHITECTURE



Human analyzes algorithm to uncover explanatory heuristics of the machine.



Human makes sense of machine output, checking for consistency and compliance, to another stakeholder.

EXAMPLE ACTIVITIES

Algorithmic testing and editing:

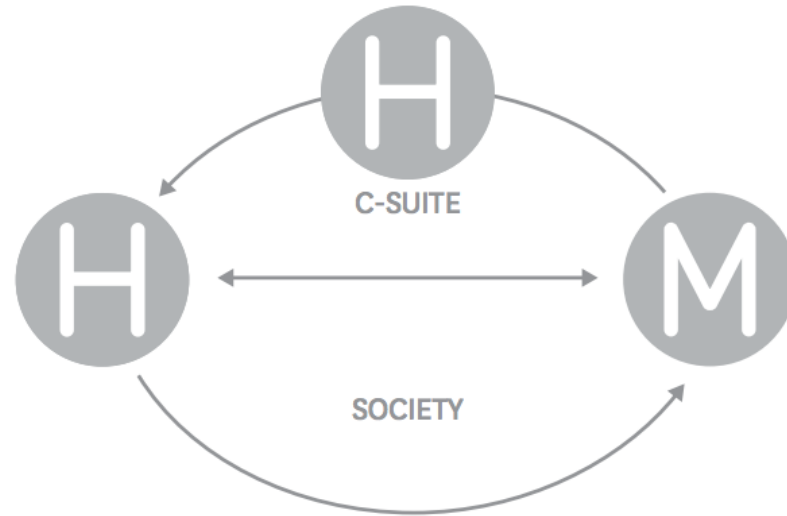
- Test, observe, and explain algorithms
- Update interface to add explainability

Output interpretation:

- Interpret machine outputs into insights
- Make sense of outputs
- Explain machine workings to stakeholders

Jobs for sustainers

RELATIONSHIP ARCHITECTURE



With C-suite input, human worker oversees the performance of the machine, imposing limits and restraints or making exceptions where necessary for sustainability and stakeholder value.

EXAMPLE ACTIVITIES

Limiting:

- Set limits or override decisions based on profitability or legal or ethical compliance

Overseeing:

- Ensure data quality, check output quality
- Apply critical thought to AI performance
- Flag errors and bad machine judgment
- Design interfaces for the AI-amplified workforce
- Manage the performance of AI by promoting, demoting, or removing a system based on societal and business impact

Responsible and sustainable AI: Emerging considerations for sustainers

Explainability	In necessary cases, use non-blackbox models so intermediate steps are interpretable and outcomes are clear, providing transparency to the process.
Accountability	Explicit identification of which decisions are delegated to machines, which decisions require human intervention, and who is accountable in either case.
Fairness	Must assure AI solutions are balanced and not biased. Need to understand why decisions are made. Need protection against data bias.
Symmetry	Must make sure that our data is an asset to us as it is to others.



Data & Simulation

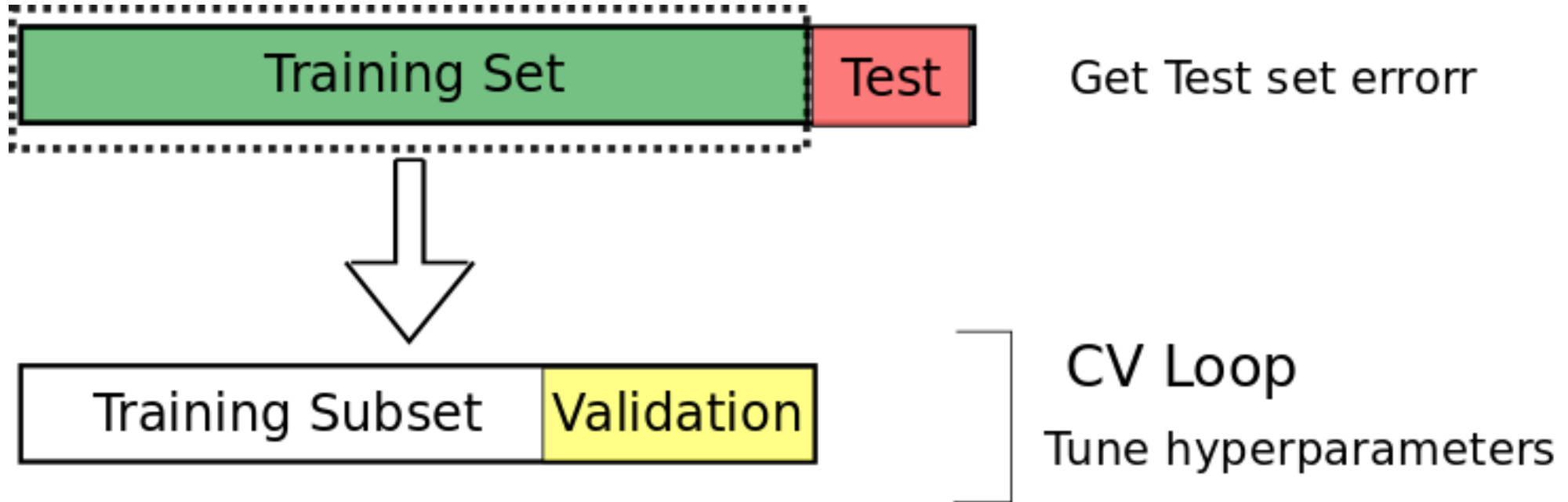
Data & Simulation Issues



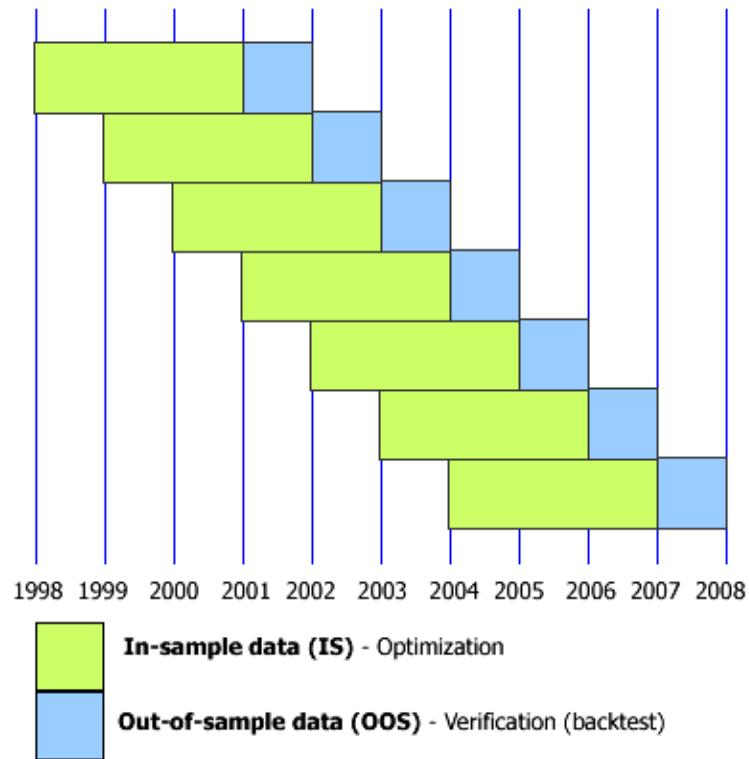
Data Processing

- Standardisation, Normalisation
- Frequency
- Windowing (time, volume, monetary turnover, 'information-based')
- Categorical data; one-hot, embeddings
- Text data (BoW)
- Log transforms (look at your input distributions)
- Non-linear transforms (box-cox, kernel trick)
- Markets without common hours
- Holidays
- Events
- Are rare events driving your returns?
- How do you reason about drawdowns?
- How representative is the data (invariance assumption)
- Feed data so invariants are clear across products

Traditional Data Partitioning

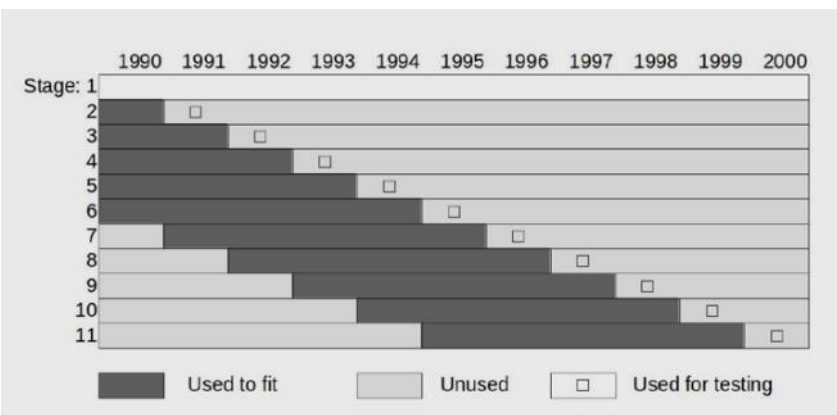
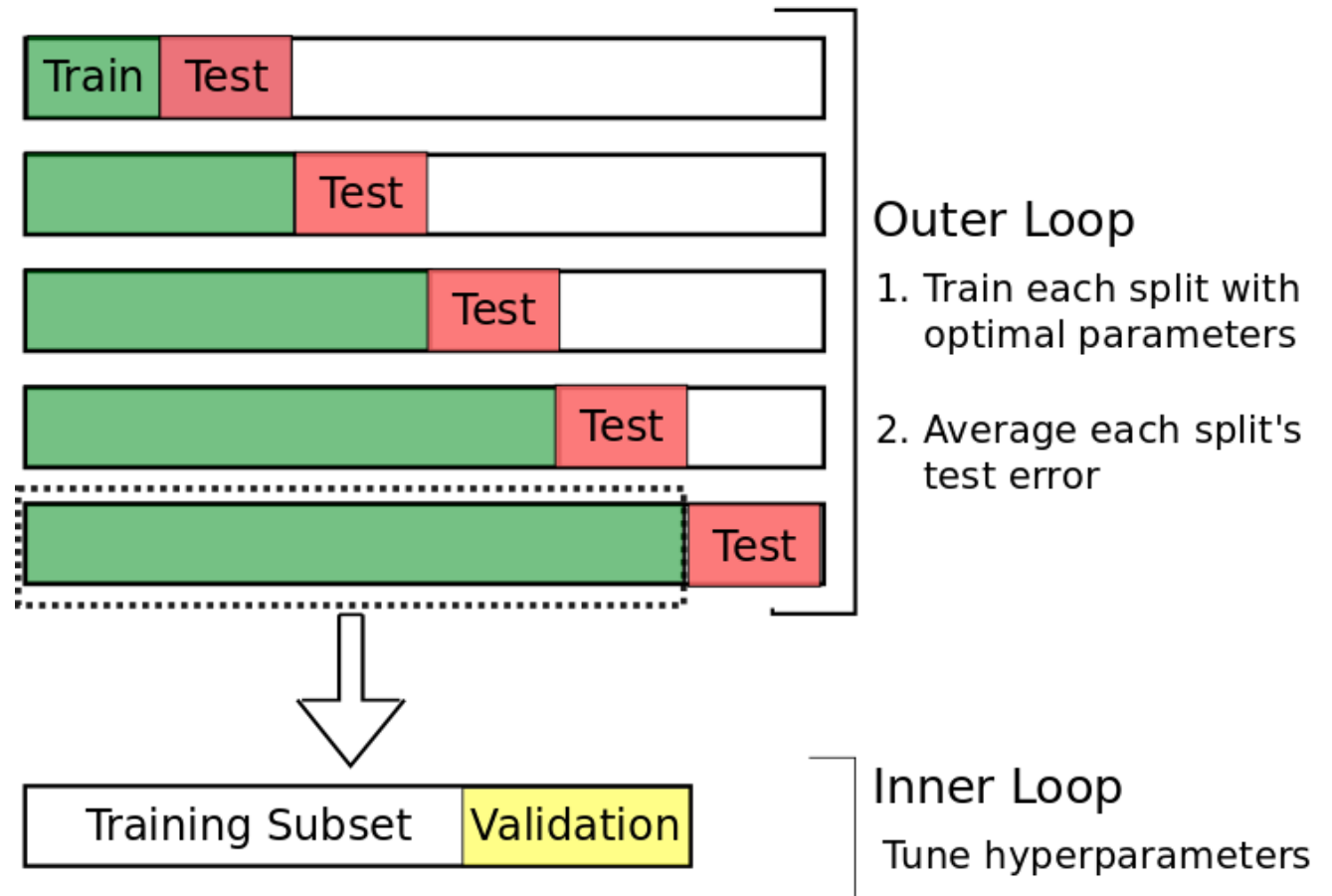


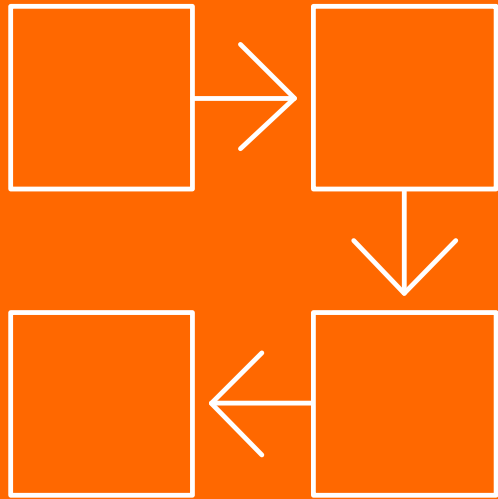
Walk-Forward Test procedure



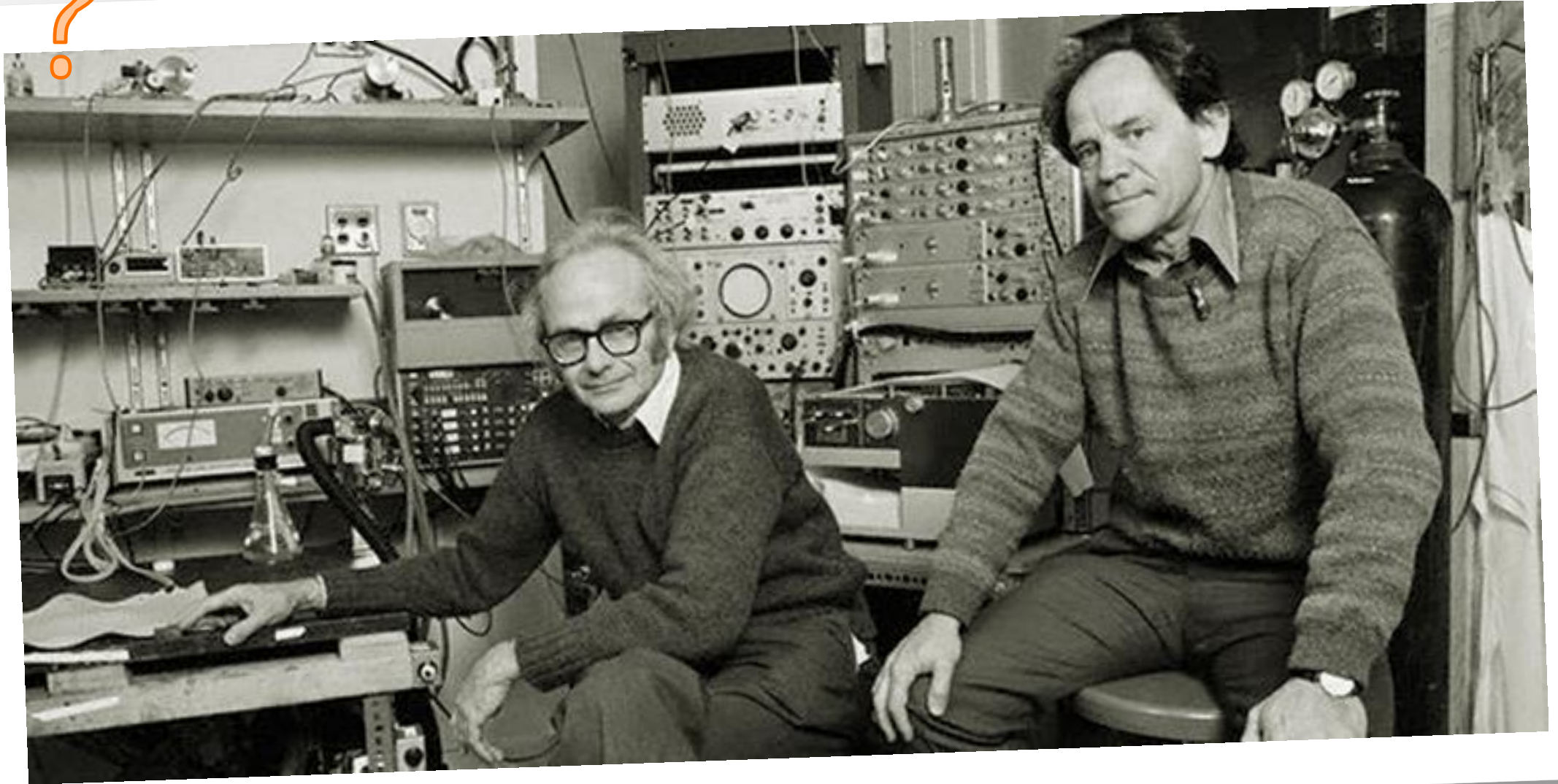
Walk Forward Testing

Nested Cross-Validation





ML & Trading



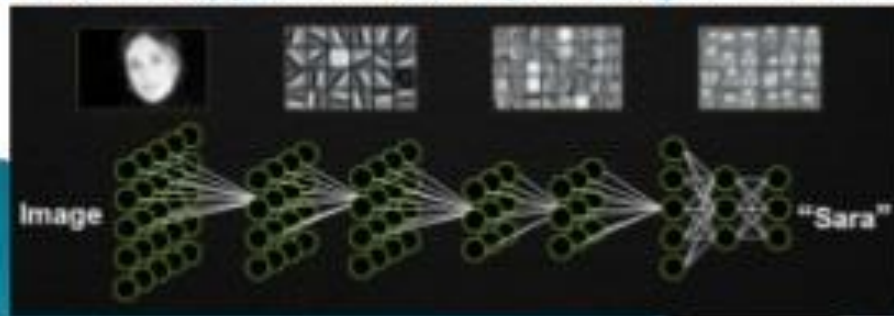
Convolutional Neural Networks

- Hubel & Wiesel inserted microscopic electrodes into the visual cortex of anesthetized cat to read activity of the single cells in visual cortex while presenting various stimuli to it's eyes during experiment on 1959. For which received noble prize under Medicine category on 1981
- Hubel & Wiesel discovered that vision is hierarchical, consists of simple cells, complex cells & hyper-complex cells

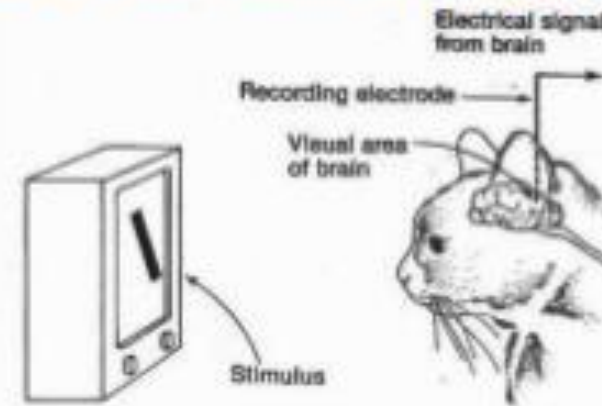
Object detection using Edges



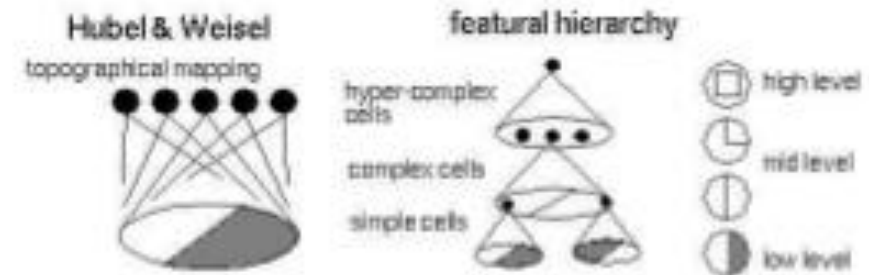
Formation of features over layers using Neural Networks



Hubel & Wiesel experiments on Cat's Vision



Vision is Hierarchical phenomenon



The Trading / World's Huber & Wiesel

- 1) What is your core domain? Do you have domain experts?
- 2) How can you work to get the machines to learn things that the human experts have learnt?
- 3) Apply the right tools for the job. Often long-standing tools are complimentary to an ML approach.

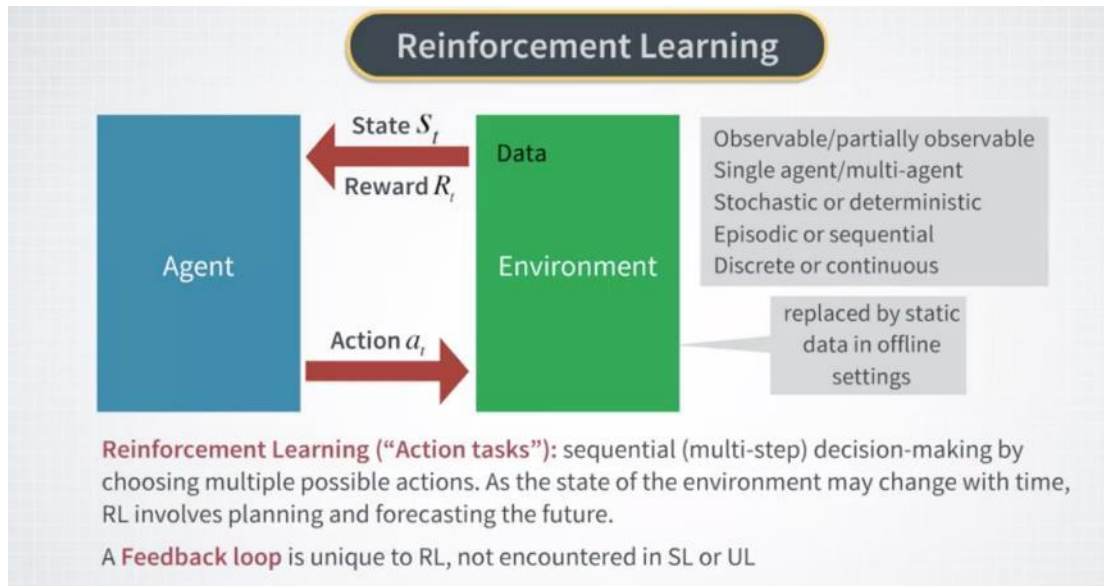
Right tool for the job?

Figure 41: Typical tasks and frequently used Machine Learning methods

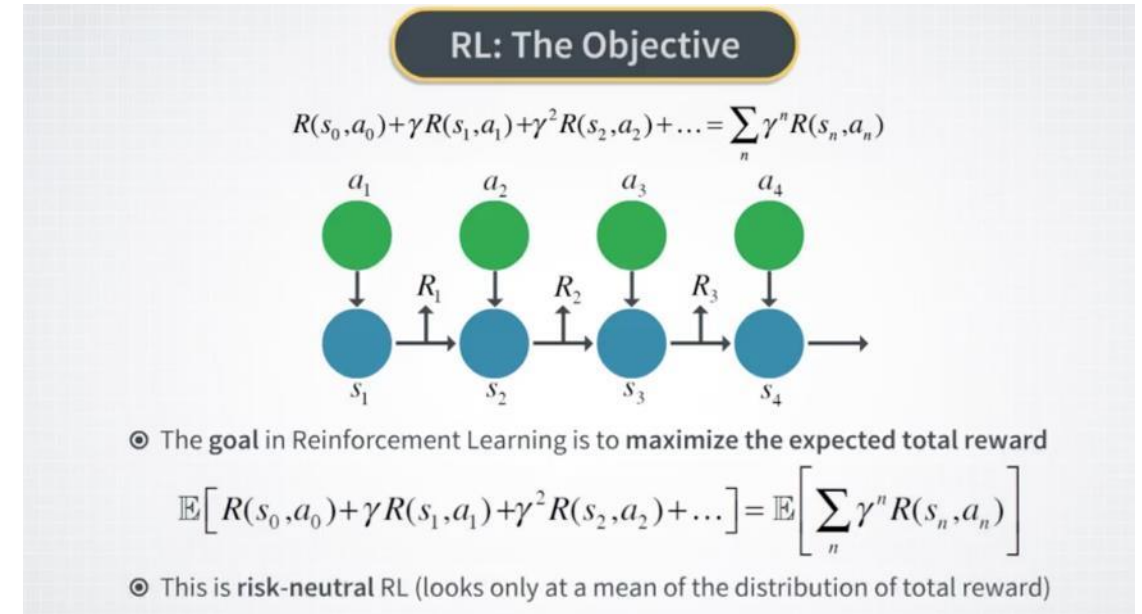
Question	Data Analysis Technique
Given set of inputs, predict asset price direction	Support Vector Classifier, Logistic Regression, Lasso Regression, etc.
How will a sharp move in one asset affect other assets?	Impulse Response Function, Granger Causality
Is an asset diverging from other related assets?	One-vs-rest classification
Which assets move together?	Affinity Propagation, Manifold Embedding
What factors are driving asset price?	Principal Component Analysis, Independent
Is the asset move excessive, and will it revert?	Component Analysis
What is the current market regime?	Soft-max classification, Hidden Markov Model
What is the probability of an event?	Decision Tree, Random Forest
What are the most common signs of market stress?	K-means clustering
Find signals in noisy data	Low-pass filters, SVM
Predict volatility based on a large number of input variables	Restricted Boltzmann Machine, SVM
What is the sentiment of an article / text?	Bag of words
What is the topic of an article/text?	Term/InverseDocument Frequency
Counting objects in an image (satellite, drone, etc)	Convolutional Neural Nets
What should be optimal execution speed?	Reinforcement Learning using Partially Observed Markov Decision Process

Source: J.P.Morgan Macro QDS

Reinforcement Learning



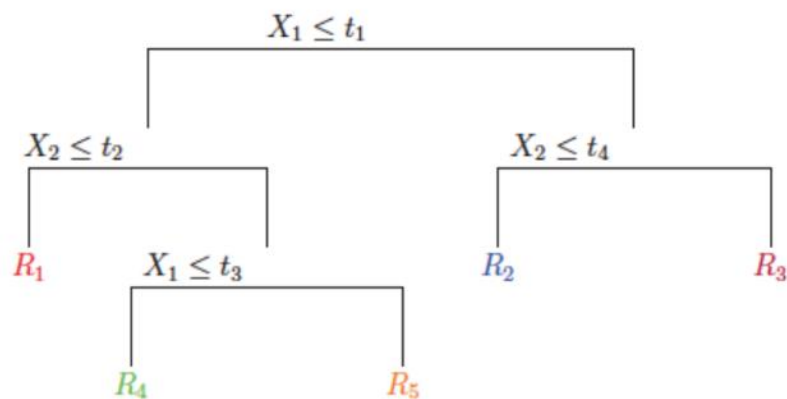
What objective?



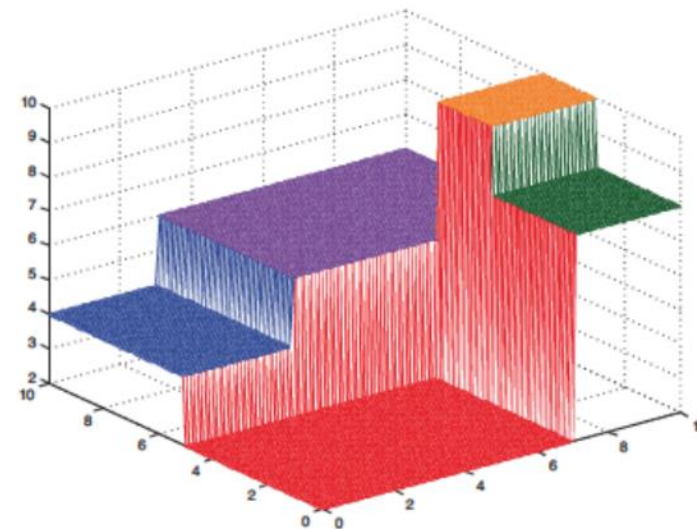
$$U_\gamma(W_t) = W_t^\gamma / \gamma \text{ for } \gamma \neq 0, \quad W_T = W_0 \prod_{t=1}^T (1 + R_t)$$

$$U_0(W_t) = \log W_t \text{ for } \gamma = 0, \quad S_T = \frac{\text{Mean}(R_t)}{\text{Std. Deviation}(R_t)}$$

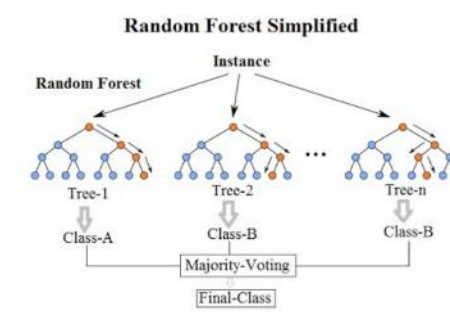
Explainable? Trees.



(a)



(b)



Explainable? Feature Importance.

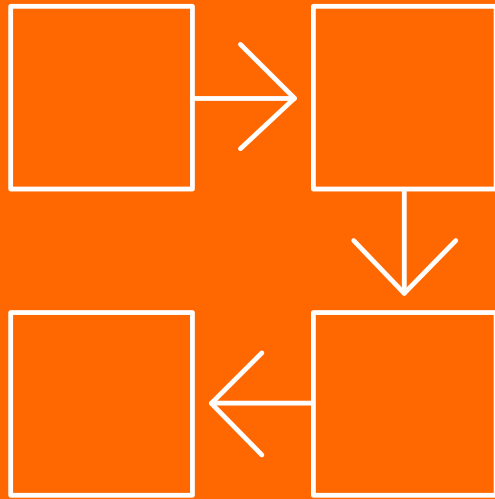
- ML algorithms identify patterns in a high dimensional space.
- These patterns associate features with outcomes.
- The nature of the relationship can be extremely complex, however we can always study what features are more important.
 - E.g., even if a ML algorithm may not derive an analytical formula for Newton's Gravitational Law, it will tell us that *mass* and *distance* are the key features.



In traditional statistical analysis, key features are often missed as a result of the model's misspecification.

In ML analysis, we give up closed-form specifications in exchange for identifying what variables are important for forecasting.

Once we know *what* are the factors at play, we can develop a theory of *how*.



Process

It is more important than ever to manage the power of a solid decision making framework. It is very easy to be fooled by randomness in financial data & using machine learning.





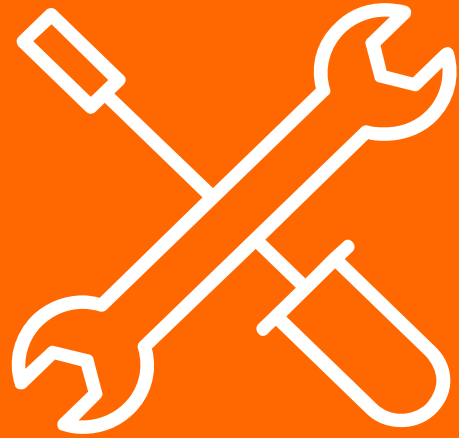
Robust Process

- What makes a decision great is not that it has a great outcome. A great decision is the result of a good process, and that process must include an attempt to accurately represent our own state of knowledge.
- Where luck is rampant, we must think of skill in terms of a process, because the results don't provide clear feedback. [Thinking in Bets]
- When luck has little influence, a good process will always have a good outcome. When a measure of luck is involved, a good process will have a good outcome but only over time.

ML Trading

- 1) Be paranoid, defensive, and assume you have made errors.
- 2) Check your assumptions about simulation relentlessly.
- 3) Invest in simulation, and in backtesting - but always be sceptical (see #1 and #2)
- 4) Watch out for selection bias everywhere.
- 5) Having data/ results in large numbers, higher frequency covers many worries. More data is the great regularizer.
- 6) Diversity beats ability. Involve a rich cross-section of people in your process to cross check and inform everything.

The first principle is that you must not fool yourself and you are the easiest person to fool.
[Richard Feynman]

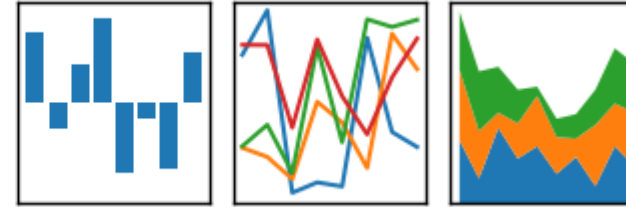


Tools



pandas

$$y_{it} = \beta' x_{it} + \mu_i + \epsilon_{it}$$



Out[84]:

	T+	T-	L+	L-	C+	C-	Ta	Tb	La	Lb	Ca	Cb
T+	0.049030	0.097990	0.057542	0.088559	0.142093	0.055787	0.124134	0.149343	0.151558	0.175534	0.104687	0.137036
T-	0.141539	0.036963	0.071593	0.032372	0.048928	0.095910	0.103252	0.135169	0.005471	0.074594	0.105789	0.141877
L+	0.008097	0.000818	0.010431	0.169142	0.004397	0.129378	0.129086	0.075599	0.131067	0.085720	0.024965	0.028616
L-	0.115121	0.164937	0.067398	0.167778	0.112021	0.023542	0.060875	0.114142	0.093648	0.104136	0.083946	0.179829
C+	0.022267	0.148255	0.132604	0.179008	0.035331	0.076943	0.016299	0.048817	0.146732	0.147461	0.101626	0.054441
C-	0.053613	0.001189	0.138876	0.021364	0.055671	0.111824	0.019809	0.050412	0.022064	0.055314	0.085893	0.048980
Ta	0.157590	0.090939	0.118423	0.002940	0.126457	0.120796	0.071413	0.010161	0.057495	0.005024	0.037983	0.017740
Tb	0.093417	0.070963	0.144675	0.076486	0.095160	0.072326	0.036108	0.127832	0.108080	0.115912	0.094242	0.035151
La	0.130178	0.167607	0.023438	0.018212	0.044306	0.030595	0.086881	0.105643	0.011486	0.102512	0.074583	0.145331
Lb	0.047221	0.102335	0.091594	0.149195	0.116416	0.127239	0.120342	0.040527	0.010295	0.009114	0.094824	0.020557
Ca	0.038426	0.114875	0.071963	0.000478	0.131425	0.017125	0.112036	0.088593	0.146312	0.081523	0.085764	0.181400
Cb	0.143501	0.003127	0.071464	0.094466	0.087796	0.138535	0.119766	0.053762	0.115792	0.043156	0.105699	0.009042

```
File Edit View Insert Cell Kernel Help
10-minute tour of pandas
from Wes McKinney
np.random.randn(1000) / 200 + 0.03,
np.random.randn(1000) / 200 + 0.08,
'ShortInterest' + np.random.randn(1000) / 300 - 0.02},
index=tickers.take(np.random.permutation(Nfull)[:1000])

df.head()

Out[276]:
Momentum ShortInterest Value
ALDGA 0.017770 -0.023712 0.063033
ARMVB 0.015964 -0.029212 0.061803
CCRDJ 0.032402 -0.028089 0.060969
ZBSPX 0.026301 -0.021976 0.060967
YRHHI 0.026148 -0.022816 0.075221

In [ ]: means = df.groupby('industries').mean()

In [ ]: means.plot(kind='bar')

In [ ]: means = df.groupby(['industries', 'ccy']).mean()
means

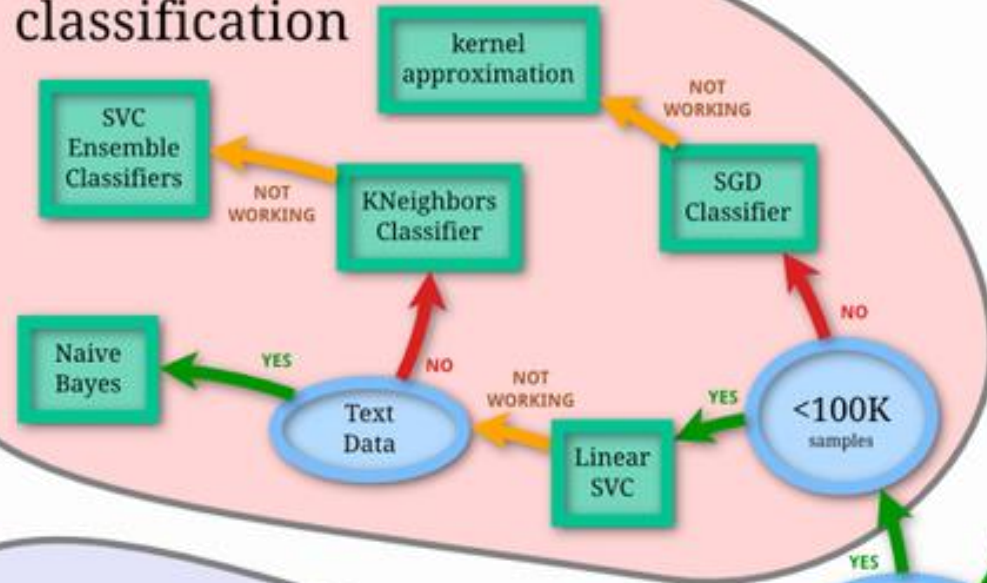
In [ ]: keys = ['industries', 'ccy']
score = (x - x.mean()) / x.std()
score.groupby(keys).agg(['mean', 'std'])

In [ ]: score.groupby(keys).agg(['mean', 'std'])
```

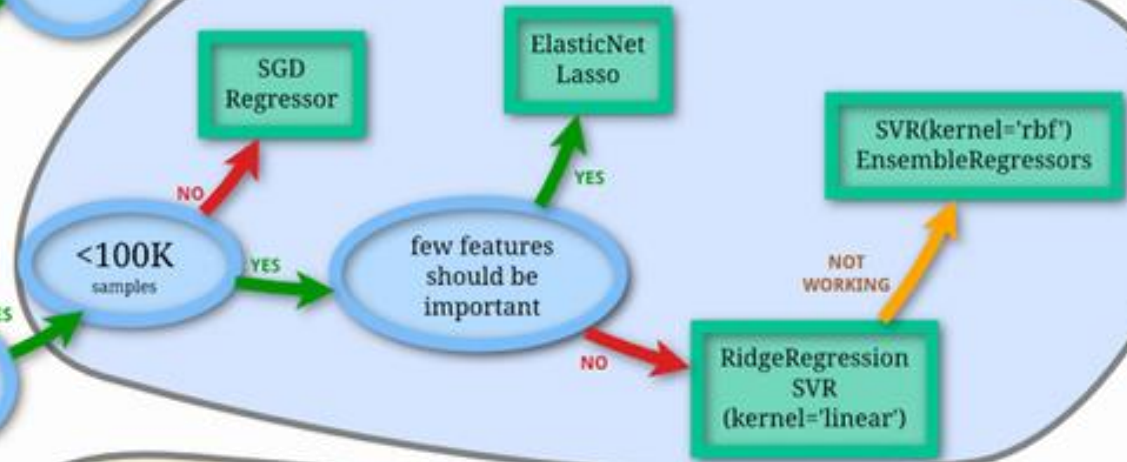
10-minute tour of pandas from [Wes McKinney](#) on [Vimeo](#).

scikit-learn algorithm cheat-sheet

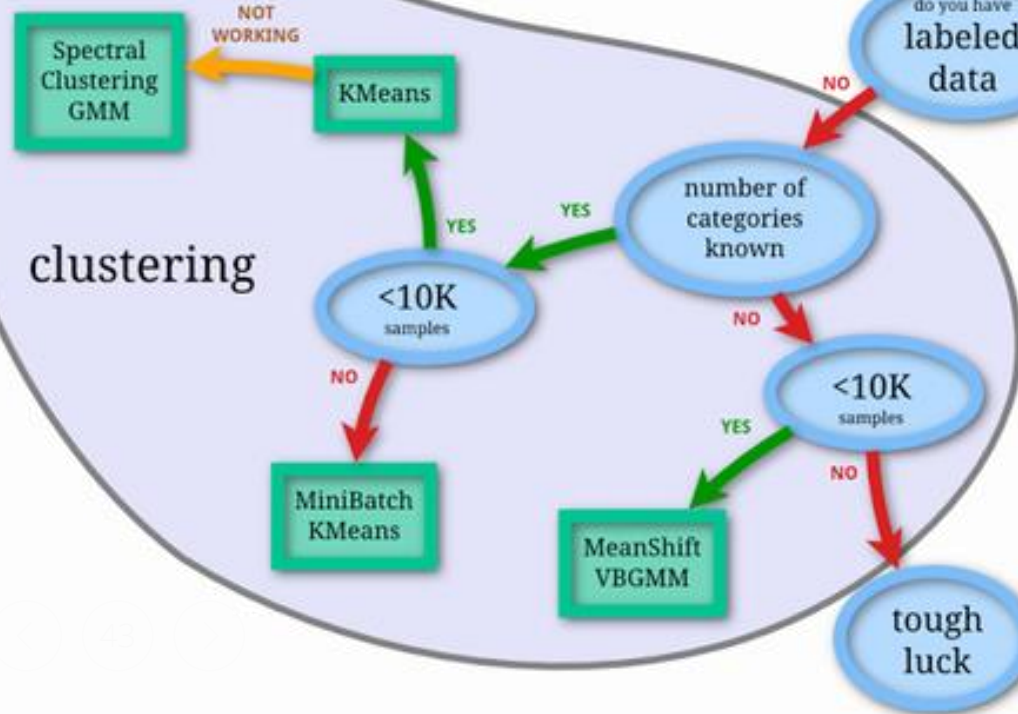
classification



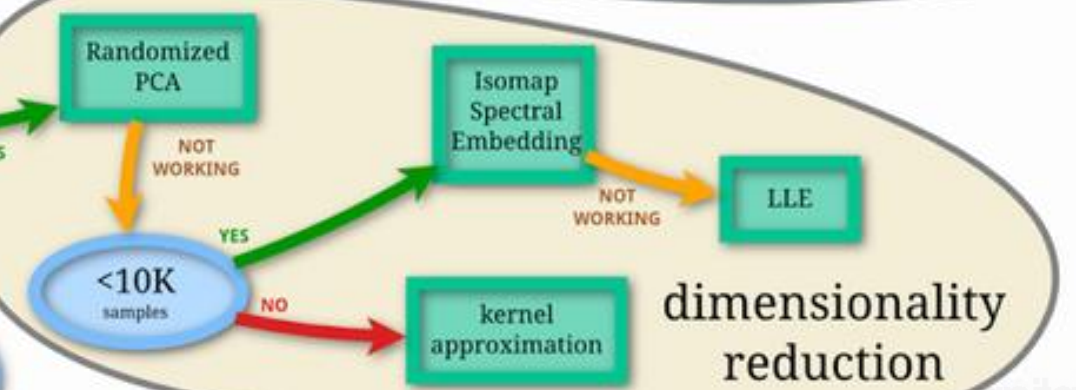
regression



clustering



dimensionality reduction





```
1 from __future__ import print_function
2 import torch
3 import torch.nn as nn
4 import torch.optim as optim
5 import numpy as np
6 import matplotlib
7 matplotlib.use('Agg')
8 import matplotlib.pyplot as plt
9
10 class Sequence(nn.Module):
11     def __init__(self):
12         super(Sequence, self).__init__()
13         self.lstm1 = nn.LSTMCell(1, 51)
14         self.lstm2 = nn.LSTMCell(51, 51)
15         self.linear = nn.Linear(51, 1)
16
17     def forward(self, input, future = 0):
18         outputs = []
19         h_t = torch.zeros(input.size(0), 51, dtype=torch.double)
20         c_t = torch.zeros(input.size(0), 51, dtype=torch.double)
21         h_t2 = torch.zeros(input.size(0), 51, dtype=torch.double)
22         c_t2 = torch.zeros(input.size(0), 51, dtype=torch.double)
23
24         for i, input_t in enumerate(input.chunk(input.size(1), dim=1)):
25             h_t, c_t = self.lstm1(input_t, (h_t, c_t))
26             h_t2, c_t2 = self.lstm2(h_t, (h_t2, c_t2))
27             output = self.linear(h_t2)
28             outputs += [output]
29         for i in range(future):# if we should predict the future
30             h_t, c_t = self.lstm1(output, (h_t, c_t))
31             h_t2, c_t2 = self.lstm2(h_t, (h_t2, c_t2))
32             output = self.linear(h_t2)
33             outputs += [output]
34         outputs = torch.stack(outputs, 1).squeeze(2)
35         return outputs
36
```

```
38 if __name__ == '__main__':
39     # set random seed to 0
40     np.random.seed(0)
41     torch.manual_seed(0)
42     # load data and make training set
43     data = torch.load('traindata.pt')
44     input = torch.from_numpy(data[3:, :-1])
45     target = torch.from_numpy(data[3:, 1:])
46     test_input = torch.from_numpy(data[:3, :-1])
47     test_target = torch.from_numpy(data[:3, 1:])
48     # build the model
49     seq = Sequence()
50     seq.double()
51     criterion = nn.MSELoss()
52     # use LBFGS as optimizer since we can load the whole data to train
53     optimizer = optim.LBFGS(seq.parameters(), lr=0.8)
54     #begin to train
55     for i in range(15):
56         print('STEP: ', i)
57         def closure():
58             optimizer.zero_grad()
59             out = seq(input)
60             loss = criterion(out, target)
61             print('loss:', loss.item())
62             loss.backward()
63             return loss
64         optimizer.step(closure)
65         # begin to predict, no need to track gradient here
66         with torch.no_grad():
67             future = 1000
68             pred = seq(test_input, future=future)
69             loss = criterion(pred[:, :-future], test_target)
70             print('test loss:', loss.item())
71             y = pred.detach().numpy()
72         # draw the result
73         plt.figure(figsize=(30,10))
74         plt.title('Predict future values for time sequences\n(Dashlines are predicted values)', fontsize=30)
75         plt.xlabel('x', fontsize=20)
76         plt.ylabel('y', fontsize=20)
77         plt.xticks(fontsize=20)
78         plt.yticks(fontsize=20)
79         def draw(yi, color):
80             plt.plot(np.arange(input.size(1)), yi[:input.size(1)], color, linewidth = 2.0)
81             plt.plot(np.arange(input.size(1), input.size(1) + future), yi[input.size(1):], color + ':', linewidth = 2.0)
82         draw(y[0], 'r')
83         draw(y[1], 'g')
84         draw(y[2], 'b')
85         plt.savefig('predict%d.pdf'%i)
86         plt.close()
```

